

CTF Writeup: (Tables) Lost in Excel Sheets

Files: Excel.xlsm + flag1.zip

Category: Office Forensics / XML Analysis

Flag: SK-CERT{L057_1N_3XC3L_5H3375}

Password Order: 1-4-2-3

1. Overview & Key Insight

The challenge involves an Excel macro file (Excel.xlsm) and a password-protected ZIP (flag1.zip). The objective is to find 4 hidden message parts and combine them in the order 1-4-2-3 to unlock the archive.

The Key Insight: Modern Office files are ZIP archives. By renaming the file to .zip or using extraction tools, we can access the raw XML files where data is stored. Excel is merely a viewer for this structured text.

2. Hiding Locations & Extraction

Part 1: Off-Screen Cells (MSG_1 = 8f2e8d95)

Location: Infrastructure sheet (xl/worksheets/sheet3.xml).

- Technique:** The hex value was split across 6 cells in row 495, columns BS through BX.
- Discovery:** These coordinates are far off-screen, requiring either massive scrolling or a simple grep of the sheet's XML.

Part 2: Deep Legacy Comments (MSG_2 = aa30c9bf)

Location: xl/comments1.xml.

- Technique:** A Note was attached to cell QXY816551.
- Discovery:** This uses an extreme row number (816,000+) and column (QXY) that are legitimate but practically unreachable via standard manual navigation.

Part 3: Obfuscated Formula Tracing (MSG_3 = 0c4810a6)

Location: Hidden sheet "Resultz" (sheet6.xml).

- Technique:** Complex CHAR(ROUND(...)) formulas referenced data on a "Very Hidden" sheet (sheet5.xml) to reconstruct characters.
- Decoding:** The formula output spelled "}6a0184c0{_3_GSM_NEDDIH". Applying a string reversal (as hinted by VBA logic) yielded HIDDEN_MSG_3_{0c4810a6}.

Part 4: Metadata Injection (MSG_4 = 83b44f09)

Location: xl/persons/person.xml.

- Technique:** The value was hidden in the userId field of the commenter metadata for "Michael Jackson".
- Discovery:** This area is pure metadata and never appears in a spreadsheet cell.

3. Final Password Assembly

Order	Part	Hex Value
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Order	Part	Hex Value
1	MSG_1	8f2e8d95
2	MSG_4	83b44f09
3	MSG_2	aa30c9bf
4	MSG_3	0c4810a6

Final Password: 8f2e8d9583b44f09aa30c9bf0c4810a6

How to Approach This Challenge (Methodology)

- 1. Extract Immediately:** Treat the Office file as a ZIP (`unzip -d output file.xlsm`).
- 2. Grep Everything:** `grep -ri "HIDDEN_MSG" .`
- 3. Check All XML:** Don't limit yourself to cell data. Check `sharedStrings.xml`, `comments1.xml`, `drawings/`, `workbook.xml`, and `docProps/app.xml`.
- 4. Audit Sheet States:** Look for `state="hidden"` or `state="veryHidden"` in `workbook.xml`.
- 5. Trace Formulas:** Search for `CHAR()`, `CODE()`, or `SUBSTITUTE()` used for obfuscation.
- 6. Inspect Extreme Addresses:** Check cells like Row 816,000 or Column QXY.

Tools Used: unzip, grep, Python (`xml.etree.ElementTree`), manual formula tracing.

Lessons Learned

- "Hidden" in Excel only means "not visible by default." Raw data remains in the XML.
- Metadata fields (Author, User ID, Properties) are prime hiding spots.
- "Very Hidden" sheets are real and require XML/VBA access.
- Complex formulas can turn numeric gibberish into legible strings.
- **Always extract and grep Office files in CTFs. Never just open them.**

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